

Samuel M. Karumba, PhD

Research Scientist | Trustworthy AI and Secure Distributed Systems

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PROFESSIONAL SUMMARY

Applied security and privacy researcher with over ten years of experience across academia, national research, and industry. My work spans GuardlyData, CSIRO Data61, CSIRO Energy, UNSW Sydney, and IBM Research Africa. I design secure and trustworthy systems for sensitive data, critical infrastructure, and multi-party coordination. My experience covers AI security, privacy-preserving computation, applied cryptography, cybersecurity, and secure distributed systems. I have built research systems, prototypes, and enterprise-facing security architectures across generative AI data-loss prevention, cyber-physical energy infrastructure, blockchain-enabled coordination, secure data sharing, and privacy-preserving analytics. My outputs include peer-reviewed publications, granted patents, technical reports, deployed prototypes, and national research program contributions. My current work focuses on AI security and privacy, including sensitive-data detection, local-first privacy controls, prompt and source-code exposure risks, shadow AI governance, and secure enterprise adoption of generative AI.

RESEARCH INTERESTS

- AI security and privacy: sensitive-data detection, secure enterprise adoption of generative AI, and real-time governance of AI interactions
- Privacy-preserving computation: zero-knowledge proofs, secure multiparty computation, and homomorphic encryption
- Security and privacy of distributed, decentralized, and cyber-physical systems
- Trust-aware coordination in distributed systems, including consensus and cross-system interoperability
- Applied cryptography for infrastructure, fintech, energy, and multi-party data-sharing systems

EDUCATION

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|---------------------|---|
| May 2019 – Feb 2024 | Doctor of Philosophy (PhD), Computer Science and Engineering
University of New South Wales (UNSW) , Sydney, Australia
UNSW Scientia Scholarship recipient.
<i>Supervisors: Prof. Salil S. Kanhere; Prof. Raja Jurdak; Dr. Subbu Sethuvenkataraman</i>
<i>Thesis: “Blockchain-based decentralized energy systems: advancing scalability, security, privacy, and interoperability.”</i> |
| Sep 2013 – Jul 2017 | Master of Science, Mobile Telecommunication Innovation and Development
Strathmore University , Nairobi, Kenya (Safaricom Academy).
Safaricom Academy Scholarship recipient.
<i>Supervisor: Dr. Vitalis Gavole Ozianyi</i>
<i>Dissertation: “Smart education: an event framework for cognitive blended learning.”</i>
GPA: 3.56 / 4.00 |
| Sep 2009 – Dec 2012 | Bachelor of Science, Computer Science
Moi University , Eldoret, Kenya.
Second Class Honours (Upper Division). |

RESEARCH AND TECHNICAL SKILLS

AI Security, Privacy, and Data Protection

- AI security and privacy for enterprise, cloud, developer, and collaboration environments.
- GenAI data leakage risk, shadow AI governance, and sensitive-data exposure analysis.
- Semantic detection of sensitive information across prompts, uploads, documents, browser activity, and developer workflows.
- Local-first privacy controls, policy enforcement, auditability, and data minimisation for AI-enabled systems.
- Security and governance requirements for safe adoption of GenAI tools in regulated and data-sensitive organisations.

Privacy-Preserving Computation and Applied Cryptography

- Secure multiparty computation, private aggregation, threshold protocols, and confidential data processing.
- Zero-knowledge proofs, selective disclosure mechanisms, and verifiable computation concepts.
- Homomorphic encryption, including Paillier-based encrypted computation for distributed data systems.
- Cryptographic protocol design for confidentiality, integrity, authentication, and trust.
- Privacy-preserving computation for AI, energy, fintech, infrastructure, and multi-party data-sharing systems.

Distributed and Blockchain Systems

- Secure distributed systems, decentralised architectures, and trust-aware coordination mechanisms.
- Blockchain security, consensus mechanisms, and permissioned and consortium blockchain systems.
- Cross-chain interoperability protocols, notary-based verification, asset transfer, and data exchange.
- Smart contract design, security analysis, and empirical evaluation of blockchain-based systems.
- Platforms and tooling: Hyperledger Fabric, Fabric-CA, chaincode, Ethereum, Solidity, and Hyperledger Caliper.

IoT, Edge, and Resource-Constrained Systems

- IoT-blockchain integration, sensor data anchoring, and secure edge gateways.
- Context-aware, mobile, embedded, and offline-first system architectures.
- Resource-constrained system design for intermittent connectivity, limited compute, and field deployment constraints.
- Mobile and device integration: Android SDK, BLE, GSM/GPRS, USSD, SMS, and sensor-driven data collection.

Systems Engineering, DevOps, and Research Tooling

- Programming languages: Go, Python, Solidity, Java, TypeScript, and JavaScript.
- Distributed backend development, APIs, microservices, gRPC, data-processing pipelines, and research prototypes.
- Containerisation and orchestration: Docker, Kubernetes, Helm, and cloud-hosted development environments.
- CI/CD and experimentation pipelines: GitHub Actions, Jenkins, reproducible evaluation, and benchmark automation.
- Cloud and infrastructure platforms: IBM Cloud, AWS IoT Core, Infura, and blockchain deployment environments.

RESEARCH EXPERIENCE

Research Scientist – AI Data Security and Privacy

Jan 2025 – Present

GuardlyData, Remote (Canada)

- Build and fine-tune NER models for sensitive-information and secret detection, optimized for on-device inference (PyTorch, Hugging Face Transformers).
- Design local-first detection, redaction, blocking, and auditing of data exposure across browsers, copilots, IDEs, RAG, and agentic AI workflows.
- Conduct threat modelling for AI data leakage, shadow AI, prompt-based exposure, source-code leakage, and insider risk.
- Translate security research into product requirements, architecture, roadmaps, and investor-facing technical documentation.

Postdoctoral Research Fellow – Cyber-Physical Systems Security

Dec 2023 – Dec 2024

CSIRO Data61, Sydney, Australia

- Conducted applied cybersecurity research within the CAESER (Consumer Energy Resources) national program.
- Led threat modelling for smart inverters and distributed energy resources, producing taxonomies and risk-assessment frameworks.
- Published peer-reviewed work (ACM e-Energy) and stakeholder reports; engaged industry via workshops and briefings.

Affiliated PhD Researcher – Secure Building Data Management

May 2020 – Dec 2023

CSIRO Energy Business Unit, Sydney, Australia

- Industry-embedded research on the Data Clearing House (DCH), a semantic building-data platform (200+ buildings, ~5 MW flexible capacity).
- Contributed to semantic modelling of heterogeneous time-series data (BMS, IoT, metering) using the Brick Schema.
- Designed privacy-preserving data-sharing and access control supporting the platform's data-sovereignty model.

PhD Research Student – Blockchain and Distributed Energy Systems

May 2020 – Dec 2023

CSIRO Energy BU, Sydney, Australia

- Doctoral research on secure, privacy-preserving distributed systems and decentralized platforms.
- Designed frameworks (HARB, BAILIF, CypherChain, PlexiChain) for privacy-preserving aggregation and cross-system interoperability.
- Published in IEEE Internet of Things Journal, IEEE ICBC, ACM e-Energy, and JNCA.

Research Software Engineer – Blockchain and Distributed Systems

Sep 2014 – May 2019

IBM Research Africa, Nairobi, Kenya

- Co-architected blockchain systems for IBM's research portfolio, including **TradeLens**, serving 100,000+ users across 600 ports.
- Led mobile and backend development integrating event frameworks, sensor analytics, and learning models.
- Built prototypes for education, agriculture, and logistics (up to 25% improvement in data-synchronization latency).
- Filed patents (US9961139B2, US20180189449A1, US11497156B2) on blockchain, IoT, and medical-data systems.

SELECTED RESEARCH PROJECTS

AI-native data protection for enterprise AI workflows

Jan 2025–Present

Research Scientist, GuardlyData

- Designed detection architectures for identifying sensitive information and risky data movement across copilots, browsers, IDEs, RAG systems, and agentic workflows.
- Built and evaluated on-device NER and secret-detection models for real-time redaction, blocking, warning, and audit workflows.
- Developed data-labeling, validation, and secure pipeline processes for PII, credentials, and sensitive-content detection.

CAESER: Cybersecurity analysis for consumer energy resources

2024

Postdoctoral Research Fellow, CSIRO Data61

- Developed threat models and security taxonomies for consumer energy resources, including smart inverters and grid-edge devices.
- Analysed attack surfaces affecting distributed energy resources, utilities, regulators, and critical energy infrastructure.
- Produced technical reporting to support stakeholder engagement and cybersecurity guidance for energy-sector decision makers.

EnergiPay: Off-chain payment channels for peer-to-peer energy trading

2023–2024

Researcher, UNSW Sydney / CSIRO Energy

- Designed an off-chain payment-channel framework for blockchain-enabled peer-to-peer energy trading.
- Modelled settlement for future energy-credit tokens to reduce on-chain transaction overhead and counterparty risk.
- **Output:** Published at ACM e-Energy 2024.

CypherChain: Privacy-preserving data aggregation for demand response

2022–2024

PhD Researcher, UNSW Sydney / CSIRO Energy

- Designed a privacy-preserving aggregation framework for blockchain-based demand response using secure multiparty computation and homomorphic encryption.
- Evaluated computation, communication, and privacy trade-offs using smart building energy data.
- **Output:** Published at IEEE ICBC 2024.

PlexiChain: Secure blockchain-based flexibility aggregation

2022

PhD Researcher, UNSW Sydney / CSIRO Energy

- Developed a blockchain-based flexibility aggregator framework for distributed energy resources and flexibility markets.
- Evaluated resilience against adversarial grid scenarios, including false data injection and coordinated demand manipulation.
- **Output:** Released as an arXiv preprint.

BAILIF: Blockchain-agnostic interoperability framework

2021–2023

PhD Researcher, UNSW Sydney / CSIRO Energy

- Designed a cross-chain interoperability framework using decentralised notaries and attestation mechanisms.
- Implemented and evaluated a proof of concept for interoperability across blockchain networks.
- **Output:** Published at IEEE ICBC 2023.

- Temporary Immutability for prosumer-side energy trading** 2020–2021
PhD Researcher, UNSW Sydney / CSIRO Energy
- Developed a removable blockchain architecture for prosumer-side energy trading where selected records can be removed under defined governance conditions.
 - Designed blackboard and removable-ledger approaches to reduce storage overhead while preserving auditability.
 - **Output:** Published in Journal of Network and Computer Applications.
- HARB: Hypergraph-based adaptive consortium blockchain** 2019–2020
PhD Researcher, UNSW Sydney / CSIRO Energy
- Developed a hypergraph-based consortium blockchain framework for scalable decentralised energy trading.
 - Designed underlay, overlay, and contract-layer models to support clustering, privacy, interoperability, and adaptive network organisation.
 - **Output:** Published in IEEE Internet of Things Journal.
- Relational network framework for distributed energy trading** 2020
PhD Researcher, UNSW Sydney / CSIRO Energy
- Designed a relational network model for analysing interoperability in distributed energy trading systems.
 - Used hypergraph-based representations to model interactions between prosumers, aggregators, markets, and blockchain infrastructure.
 - **Output:** Published as a short paper at IEEE ICBC 2020.
- Distributed education data platform** 2016–2017
Research Engineer, IBM Research – Africa
- Designed and deployed education data systems for school census and school information management in resource-constrained environments.
 - Contributed to data collection, validation, fault-tolerant ingestion, and reporting workflows for large-scale education datasets.
 - **Output:** School Census Hub and School Information Hub publications at IEEE ICSE-SEIS 2017 and ACM ICTD 2017.
- Cooperative mobile content distribution system** 2015–2016
Research Engineer, IBM Research – Africa
- Developed cooperative downloading mechanisms for resource-constrained mobile devices with limited bandwidth, battery, and storage.
 - Designed scheduling logic using device resource levels, reliability indicators, and segment allocation strategies.
 - **Output:** Published at ACM MOBILESoft 2016 and granted as US Patent US9961139B2.
- Farm cultivation quality analytics** 2017–2018
Research Engineer, IBM Research – Africa
- Contributed to a knowledge-graph and machine-learning system for assessing agricultural cultivation quality.
 - Supported modelling of farm, crop, and cultivation data for evidence-based agricultural decision support.
 - **Output:** Granted as US Patent US11497156B2.

Conference Papers

- [1] X. Hui, **S. Karumba**, S. C.-K. Chau, and M. Ahmed, “Destabilizing power grid and energy market by cyberattacks on smart inverters,” in *Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems (e-Energy '25)*, **Cited by: 1** — Conference Ranking: A-rank (CORE 2023), Association for Computing Machinery, 2025.
- [2] S. C.-K. Chau, N. Wang, and **S. Karumba**, “Energipay: Off-chain payment channel for blockchain-enabled peer-to-peer energy trading,” in *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems (e-Energy '24)*, **Cited by: 3** — Conference Ranking: A-rank (CORE 2023), Singapore, Singapore: Association for Computing Machinery, 2024, ISBN: 979-8-4007-0480-2. DOI: [10.1145/3632775.3661985](https://doi.org/10.1145/3632775.3661985).
- [3] **S. Karumba**, S. C.-K. Chau, H. Pearce, M. Ahmed, and H. Janicke, “Systematic study of cybersecurity threats for smart inverters,” in *Proceedings of the 15th ACM International Conference on Future and Sustainable Energy Systems (e-Energy '24)*, **Cited by: 3** — Conference Ranking: A-rank (CORE 2023), Singapore, Singapore: Association for Computing Machinery, 2024, pp. 669–675, ISBN: 979-8-4007-0480-2. DOI: [10.1145/3632775.3661994](https://doi.org/10.1145/3632775.3661994).
- [4] **S. Karumba**, V. Dedeoglu, R. Jurdak, and S. S. Kanhere, “Cypherchain: A privacy-preserving data aggregation framework for blockchain-based dr programs,” in *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, Conference Ranking: SJR: 0.340, Impact Factor: 3.04, IEEE, 2024.
- [5] **S. Karumba**, R. Jurdak, S. S. Kanhere, and S. Sethuvenkatraman, “Bailif: A blockchain agnostic interoperability framework,” in *2023 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, **Cited by: 15** — Conference Ranking: SJR: 0.340, Impact Factor: 3.04, h-index: 14, IEEE, 2023, pp. 1–9. DOI: [10.1109/ICBC56567.2023.10174967](https://doi.org/10.1109/ICBC56567.2023.10174967).
- [6] **S. Karumba**, S. S. Kanhere, and R. Jurdak, “A relational network framework for interoperability in distributed energy trading,” in *2020 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, **Cited by: 2** — Conference Ranking: SJR: 0.340, Impact Factor: 3.04, h-index: 14, IEEE, 2020, pp. 1–3. DOI: [10.1109/ICBC48266.2020.9169462](https://doi.org/10.1109/ICBC48266.2020.9169462).
- [7] N. Bore, **S. Karumba**, J. Mutahi, et al., “Towards blockchain-enabled school information hub,” in *Proceedings of the Ninth International Conference on Information and Communication Technologies and Development (ICTD '17)*, **Cited by: 111** — Conference Ranking: C-rank (CORE 2018), Lahore, Pakistan: Association for Computing Machinery, 2017. DOI: [10.1145/3136560.3136584](https://doi.org/10.1145/3136560.3136584).
- [8] A. Kinai, J. Mutahi, N. Bore, **S. Karumba**, et al., “Deploying large scale school census hub: An experience report,” in *2017 IEEE/ACM 39th International Conference on Software Engineering: Software Engineering in Society Track (ICSE-SEIS)*, **Cited by: 3** — Conference Ranking: A*-rank (CORE 2023), IEEE, 2017, pp. 47–56. DOI: [10.1109/ICSE-SEIS.2017.7](https://doi.org/10.1109/ICSE-SEIS.2017.7).
- [9] D. Kaguma, **S. Karumba**, A. Walcott-Bryant, and K. Weldemariam, “Towards cooperative content downloading for resource-constrained mobile devices,” in *Proceedings of the International Conference on Mobile Software Engineering and Systems (MOBILESoft '16)*, **Cited by: 1**, Austin, TX, USA: Association for Computing Machinery, 2016, pp. 189–198. DOI: [10.1145/2897073.2897083](https://doi.org/10.1145/2897073.2897083).

Journal Articles

- [10] **S. Karumba**, S. Sethuvenkatraman, V. Dedeoglu, R. Jurdak, and S. S. Kanhere, “Barriers to blockchain-based decentralised energy trading: A systematic review,” *International Journal of Sustainable Energy*, vol. 42, no. 1, pp. 41–71, Feb. 2023, **Cited by: 49 — Q2 in Energy** — Impact Factor: 3.5 — h-index: 42. DOI: [10.1080/14786451.2023.2171417](https://doi.org/10.1080/14786451.2023.2171417).
- [11] **S. Karumba**, S. S. Kanhere, R. Jurdak, and S. Sethuvenkatraman, “Harb: A hypergraph-based adaptive consortium blockchain for decentralized energy trading,” *IEEE Internet of Things Journal*, vol. 9, no. 16, pp. 14 216–14 227, 2022, **Cited by: 34 — Q1 in Computer Networks** — Impact Factor: 8.2 — CiteScore: 17.6 — h-index: 179. DOI: [10.1109/JIOT.2020.3022045](https://doi.org/10.1109/JIOT.2020.3022045).
- [12] A. Dorri, F. Luo, **S. Karumba**, S. S. Kanhere, R. Jurdak, and Z. Y. Dong, “Temporary immutability: A removable blockchain solution for prosumer-side energy trading,” *Journal of Network and Computer Applications*, vol. 180, p. 103 018, 2021, **Cited by: 28 — Q1 in Computer Networks** — Impact Factor: 7.7 — CiteScore: 21.5 — h-index: 141. DOI: [10.1016/j.jnca.2021.103018](https://doi.org/10.1016/j.jnca.2021.103018).

Patents

- [13] J. Mutahi, D. W. Kaguma, **S. M. Karumba**, et al., “Farm cultivation quality,” Patent US11497156B2, Nov. 2022. [Online]. Available: <https://patents.google.com/patent/US11497156B2/en>.
- [14] **S. M. Karumba**, C. A. Pickover, and K. Weldemariam, “Tracking items used for providing medical services,” Patent Application US20180189449A1, Jul. 2018. [Online]. Available: <https://patents.google.com/patent/US20180189449A1/en>.
- [15] D. W. Kaguma, **S. Karumba**, and K. Weldemariam, “Cooperative download among low-end devices under resource-constrained environment,” Patent US9961139B2, May 2018. [Online]. Available: <https://patents.google.com/patent/US9961139B2/en>.

Book Chapters

- [16] **S. Karumba**, V. Dedeoglu, A. Dorri, R. Jurdak, and S. S. Kanhere, “Utilizing blockchain as a citizen-utility for future smart grids,” in *Wireless Blockchain: Principles, Technologies, and Applications*, **Cited by: 4**, John Wiley & Sons, 2021, pp. 201–224. DOI: [10.1002/9781119790839.ch9](https://doi.org/10.1002/9781119790839.ch9).

Technical Reports & Software Frameworks

- [17] **S. Karumba**, S. S. Kanhere, R. Jurdak, and S. Sethuvenkatraman, *Plexichain: A secure blockchain-based flexibility aggregator framework*, arXiv preprint arXiv:2212.09064, **Cited by: 1**, 2022. arXiv: [2212.09064](https://arxiv.org/abs/2212.09064) [cs.CR]. [Online]. Available: <https://arxiv.org/abs/2212.09064>.
- [18] X. Hui, **S. Karumba**, S. C.-K. Chau, and M. Ahmed, *Destabilizing power grid and energy market by cyberattacks on smart inverters (extended version)*, arXiv preprint arXiv:2505.14175, Extended version of the paper published in ACM e-Energy ’25, 2025. arXiv: [2505.14175](https://arxiv.org/abs/2505.14175) [cs.CR]. [Online]. Available: <https://arxiv.org/abs/2505.14175>.

RESEARCH GRANTS AND FUNDING

2023 – 2024	CAESER: Cybersecurity of Consumer Energy Resources CSIRO Data61; University of New South Wales; Edith Cowan University Role: Postdoctoral Research Engineer	<i>National research program</i>
2020 – 2024	CSIRO PhD Top-Up Scholarship CSIRO Data61 Role: PhD Scholar	<i>AUD 40,000</i>
2019 – 2024	UNSW Scientia PhD Scholarship University of New South Wales Role: Principal Recipient	<i>AUD 205,000+</i>
2015 – 2018	Educational Data Analytics Initiative (Project Lucy) IBM Research Africa; RTI International; Mombasa County Role: Co-Investigator	<i>USAID / IBM</i>
2023	ACE-SIP Summer School Travel Award Algorand Centre of Excellence; Monash University Role: Research Engineer	<i>Professional development</i>

SCHOLARSHIPS & AWARDS

03 Feb 2023	ACE Summer School Stipend — Monash Univ. (Melbourne). <i>Stipend support for participation in the African Cybersecurity and Energy Systems Summer School.</i>
01 Apr 2020	CSIRO PhD Top-up Scholarship — UNSW/CSIRO Energy (Sydney). <i>Research scholarship for doctoral work on secure distributed energy systems.</i>
03 Aug 2018	UNSW Scientia Scholarship — Ph.D., UNSW (Sydney). <i>Flagship doctoral award for exceptional research potential across disciplines.</i>
22 Aug 2013	Safaricom Academy Scholarship — M.Sc. Mobile Telecommunication & Innovation, Strathmore Univ. (Nairobi). <i>Industry-backed full scholarship delivered under the Safaricom–Strathmore partnership for advanced mobile innovation training.</i>

REFEREES

Prof. Salil S. Kanhere Relation: Primary PhD supervisor (2019-2024) Contact: available upon request Research Areas: IoT, Blockchain, Cyber-Physical Systems, Cybersecurity, Applied ML	Professor, UNSW, Australia
Prof. Raja Jurdak Relation: PhD co-supervisor (2019-2024), Director of Trusted Networks Lab Contact: available upon request Research Areas: IoT, Blockchain, Distributed Systems, Energy Systems	Professor, QUT, Australia
Dr. Sid Chi-Kin Chau Relation: Research manager at CSIRO Data61 (2023-2024) Contact: available upon request Research Areas: Cybersecurity for Energy Systems, Blockchain, IoT Security	Principal Research Scientist, CSIRO Data61